Muhammad Ahzam

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EDUCATION

Institute of Business Administration (IBA), Karachi Expected 2027
BS in Computer Science
Highlighted Coursework: Machine Learning, Natural Language Processing with Deep Learning, Statistical Inference, Linear Algebra, Numerical Analysis, Discrete Math, Data Structures, Design and Analysis of Algorithms, Database Systems, Cloud Computing, Operating Systems, Computer Architecture and Assembly Language, OOP, Software Engineering, Web-Based App Development

EXPERIENCE

AI-Integrated Mobile App Developer Intern, EdgeFirm Jul 15 – Aug 31, 2026

- Took over the Weather Ways project mid-cycle and delivered its remaining three sprints on a live three-repository stack (Flutter/GetX, Node.js/Prisma, Next.js) through client User Acceptance Testing.
- Integrated the Google Gemini API to turn real-time weather context into dynamic route optimizations, managing app state and latency so AI output became safe, navigable routes.
- Added AWS CloudWatch (backend) and Firebase Crashlytics (mobile) to catch silent failures and edge cases in production.
- Fixed client-reported map anomalies by rewriting coordinate-parsing logic, so backend spatial data renders as smooth traffic-flow polylines.

Math & Problem-Solving Instructor, IBAGrads (IBA and SAT entry-test prep institute), Karachi Jul 2024 – Apr 2026

- Taught about 70 students across four batches for the IBA and SAT entry test; selected after ranking 1st in the IBA entry test. Students are now enrolled across IBA programs and abroad universities. plus 4 private O-level students.

PROJECTS

Multi-Agent Path Finding with Priority-Based Routing (solo) | *Python, PyTorch, DQN*

- Built a decentralized path-finding engine for robots on a 25×35 hospital pharmacy grid with 1-tile storage aisles and 2–3 tile corridors, combining A* search with a Constraint Satisfaction framework. Moves and 90° turns each cost one time unit.
- Resolved deadlocks with Priority Inheritance with Backtracking, enforcing yielding behavior in single-tile aisles.
- Trained a Deep Q-Network with curriculum learning, a 10,000-sample experience replay buffer and a custom reward balancing progress, collisions and time-based urgency.
- Achieved 100% success and a 1.0602 cost-efficiency ratio across six deterministic edge-case scenarios, eliminating livelock oscillations.

Hybrid RAG System for Forensic Science Queries (2-person team, equal contributors) | *Python, LangChain, Pinecone, Sentence Transformers*

- Built a RAG pipeline answering questions on fingerprints, blood groups and dermatoglyphics strictly from a specialized PDF corpus, with recursive chunking and a Qwen LLM via OpenRouter.
- Combined Pinecone semantic search with local BM25 keyword search, merged with Reciprocal Rank Fusion to balance semantic understanding with lexical precision.
- Designed a custom evaluation framework scoring faithfulness (word overlap) and relevancy (cosine similarity).

– Reduced response latency by 35% to 1.544 s while maintaining accuracy.

PantryPal: Meal Planning Platform (course project, 3 members; Lead Architect & Developer) | *React, Node.js, PostgreSQL*

– Architected a full-stack pantry and recipe app with PL/pgSQL stored procedures for real-time pantry deductions and daily macro aggregation.

SKILLS

– **ML/AI:** PyTorch, TensorFlow, scikit-learn, LangChain, Sentence Transformers, Pinecone, BM25, Reinforcement Learning (DQN), RAG, MLflow, Gemini API

– **Data:** Python, NumPy, Pandas, SQL, PostgreSQL

– **Languages & Web:** C++, JavaScript, React, Node.js

– **Tools:** Git/GitHub, VS Code, Cursor, Claude Code, Firebase, AWS CloudWatch, MS Excel

LEADERSHIP

Module Head, Enigma, IBA's flagship event 2023, 2024

– Selected as module head two years running; led a 20-person team running a module for 16 competing teams (144 participants).

CERTIFICATIONS

– Supervised Learning with scikit-learn – DataCamp

– Cluster Analysis in Python – DataCamp

– Introduction to MLflow – DataCamp

– AI for Professionals – LUMSX/ILMx

– Introduction to Git – DataCamp (In Progress)

– Business Communication – ILMx